Batterjee Medical College
For Science and Technology

BMC

Cardiomyopathies

An approach to a Case of Dyspnea

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Learning Objectives

Knowledge & understanding

- Discuss the etiology of dilated, hypertrophic, and restrictive cardiomyopathy.
- Explain the pathophysiologic mechanisms underlying dysfunction in dilated, hypertrophic, and restrictive cardiomyopathy.

Skill

- Recognize clinical features that warrant referral for implantable cardioverter-defibrillator placement in a patient with cardiomyopathy at risk of sudden cardiac death.
- Apply evidence-based management plan for dilated, hypertrophic, and restrictive cardiomyopathy.

Main Subtopics

- Overview & Classification of Cardiomyopathies

  (definition; dilated, hypertrophic, and restrictive types)
- Dilated Cardiomyopathy (epidemiology; etiology — ABCCCDD mnemonic; pathophysiology; clinical features; diagnostics; management)
- Hypertrophic Cardiomyopathy (epidemiology; genetics and etiology; pathophysiology — obstructive vs. non-obstructive; clinical features; physical examination findings; diagnostics; management)
- Restrictive Cardiomyopathy (epidemiology; etiology — PLEASeHelp mnemonic; pathophysiology; clinical features; diagnostics; management)
- Differentiating Cardiomyopathies by Echocardiographic Findings (LV cavity size, ejection fraction, wall thickness)

Case scenario

A 48-year-old woman comes to the physician because of progressively worsening dyspnea on exertion and fatigue for the past 2 months. She has no history of serious illness and takes no medications. Her father died from complications related to amyloidosis. She does not smoke or drink alcohol. Her temperature is 36.7°C (98°F), pulse is 124/min, respirations are 20/min, and blood pressure is 98/60 mm Hg. Cardiac examination shows no murmurs. Coarse crackles are heard at the lung bases bilaterally. An ECG shows an irregularly irregular rhythm with absent P waves. An x-ray of the chest shows globular enlargement of the cardiac shadow with prominent hila and bilateral fluffy infiltrates. Transthoracic echocardiography shows a dilated left ventricle with an ejection fraction of 40%.

Case scenario

Which of the following is the most likely cause of this patient's condition?

a. Amyloid deposition
b. Acute psychological stress
c. Thiamine deficiency
d. Chronic tachycardia
e. Coronary artery occlusion

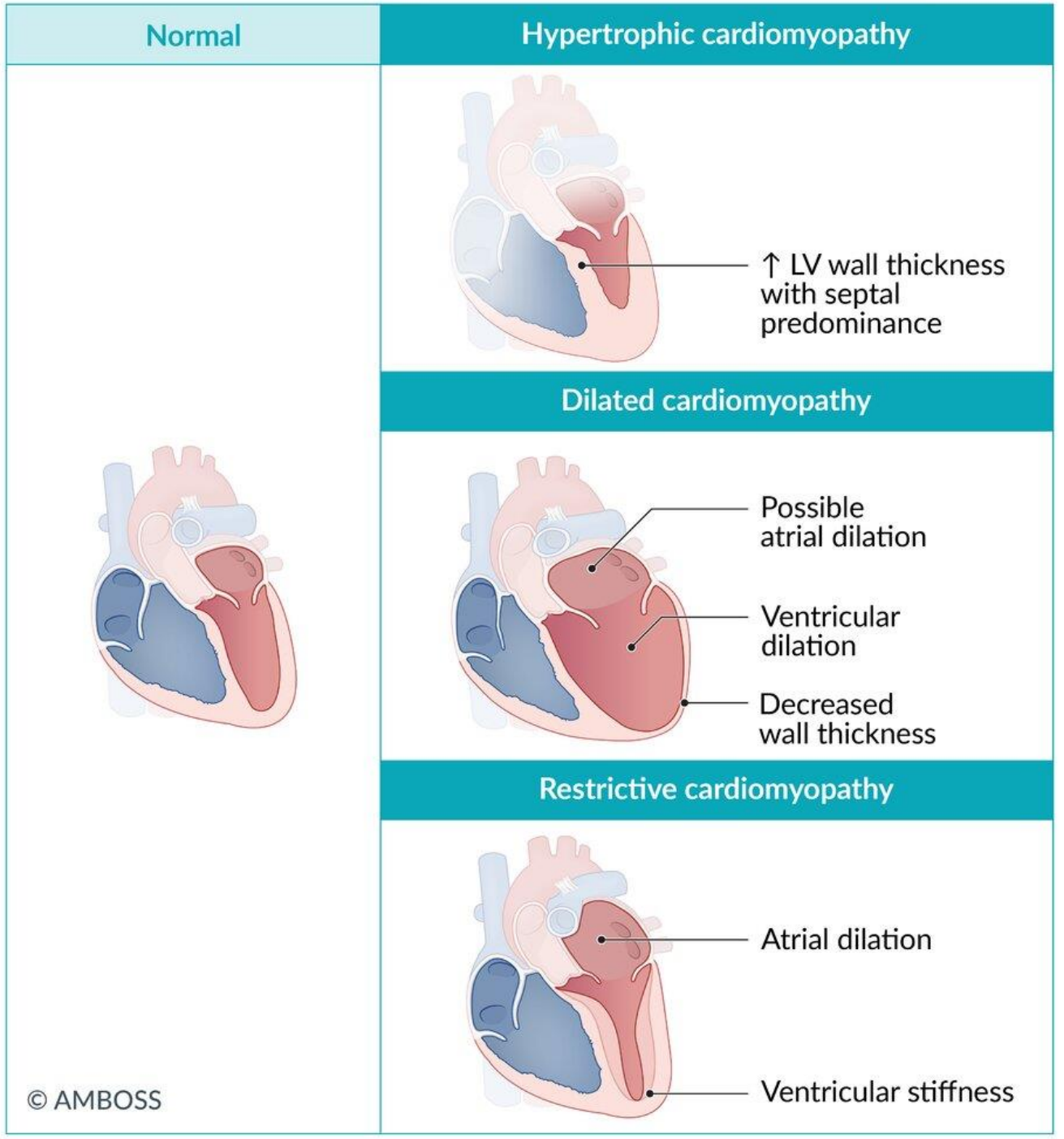
Overview

Definition:

- Cardiomyopathies are diseases of the muscle tissue of the heart
- Occur in the absence of pressure overload, volume overload, or coronary artery disease

Types:

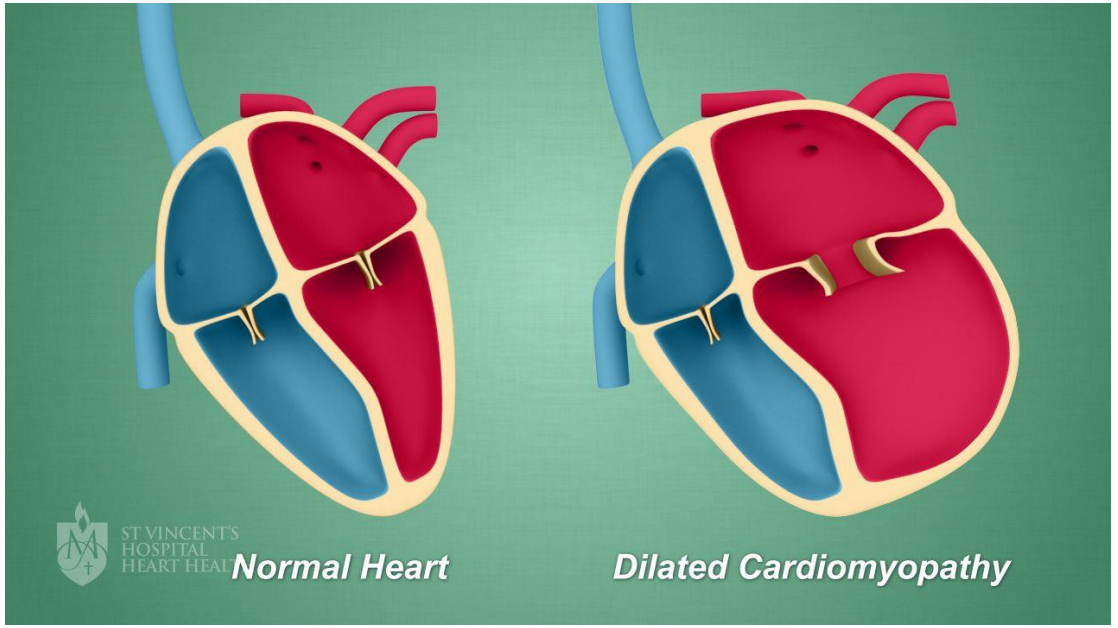
- Dilated
- Hypertrophic
- Restrictive



Dilated Cardiomyopathy (DCM)

Definition:

- A reduction in the contractile function of the ventricles
- Results in cardiac enlargement and systolic dysfunction

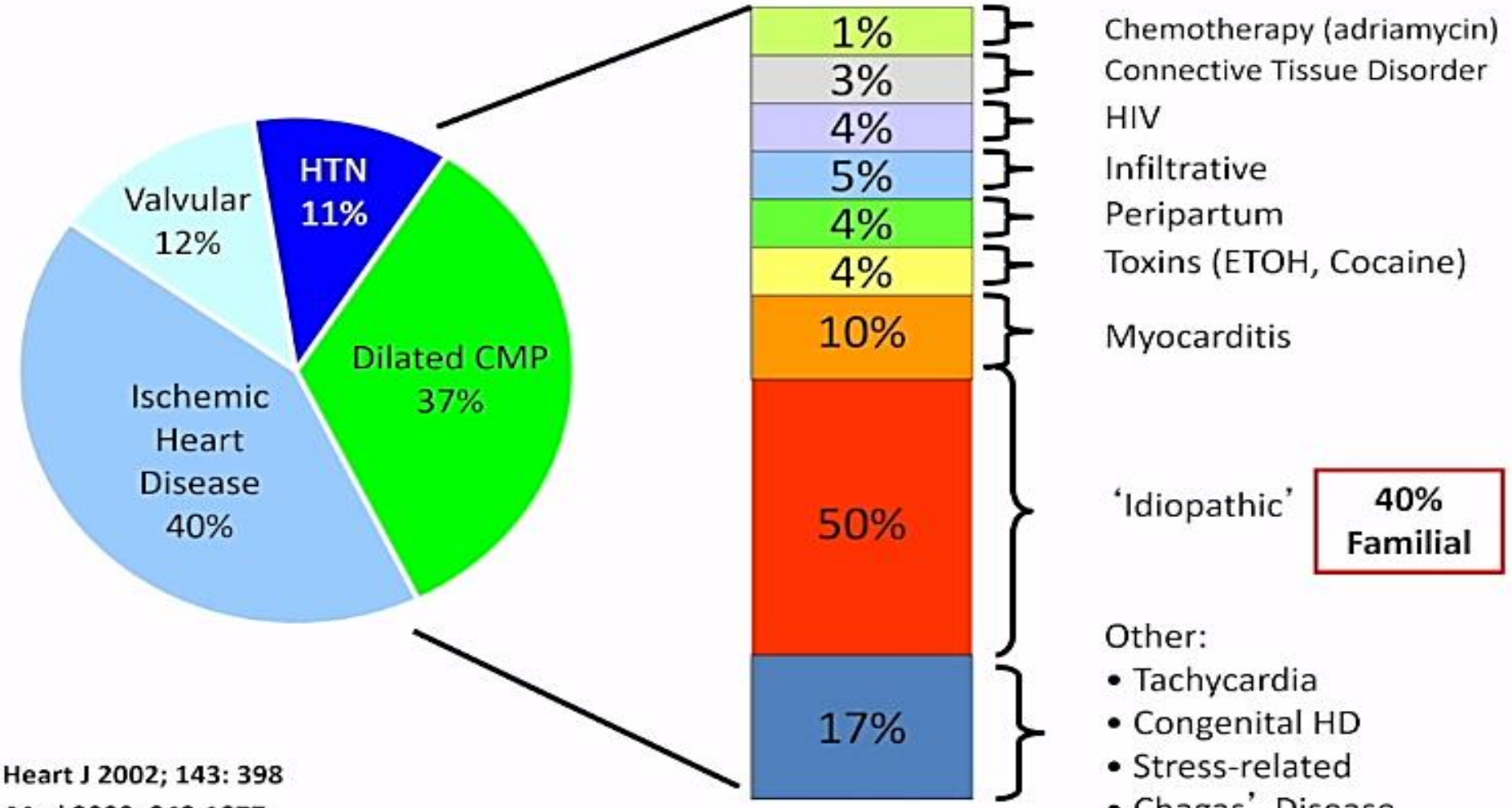


Epidemiology:

- Incidence: 6/100,000 per year (Most common cardiomyopathy)
- ♂ > ♀ (Approx. 3:1)

Dilated Cardiomyopathy (DCM)

Etiology:



aldasseroni, Am Heart J 2002; 143: 398

elker, New Engl J Med 2000; 342:1077

Dilated Cardiomyopathy (DCM)

Secondary causes:

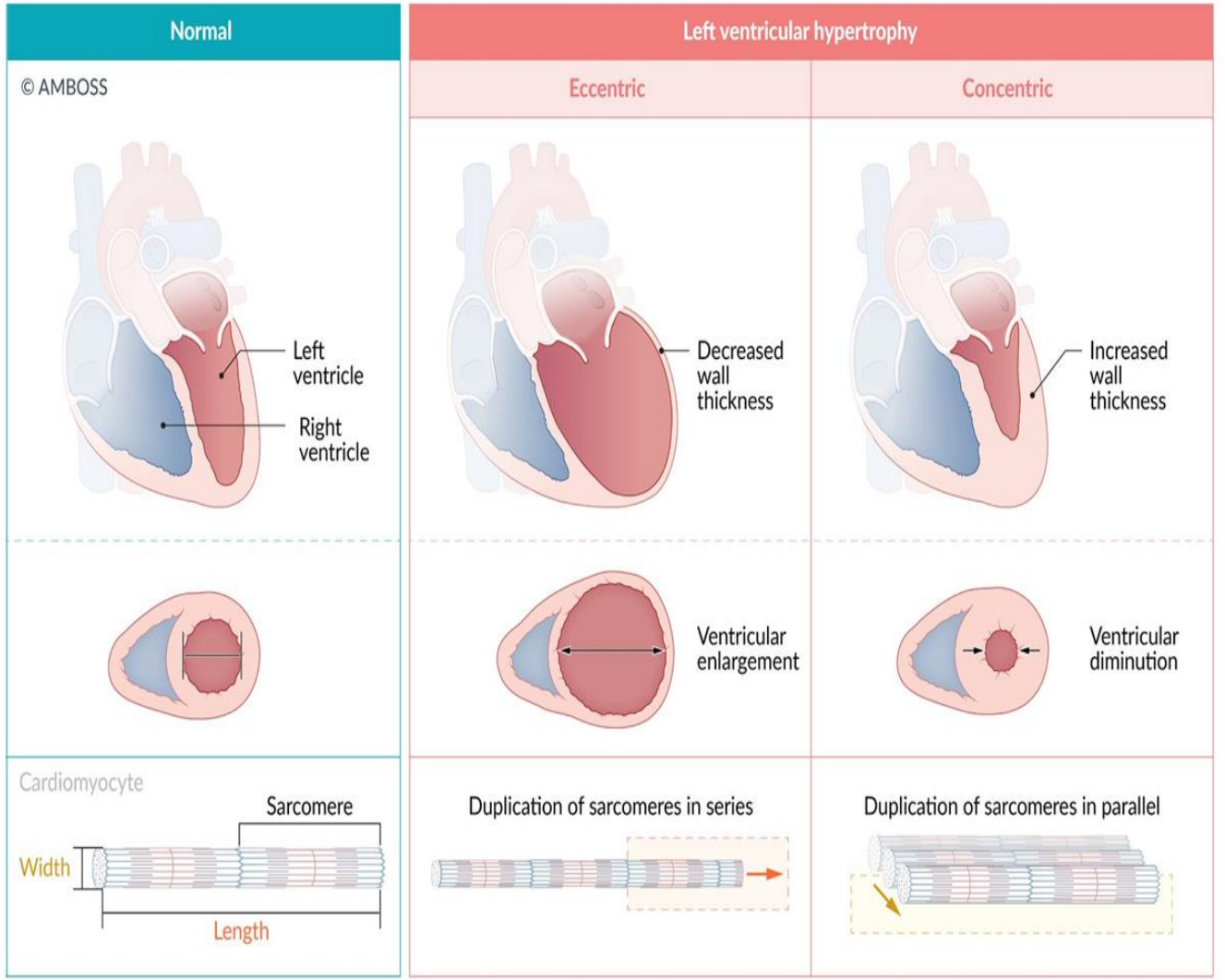
- Peripartum cardiomyopathy
- Infection (Coxsackie B virus, Chagas disease)
- Sarcoidosis
- Radiation and Chemotherapy
- Toxic substances
- Malnutrition (Thiamine deficiency)
- Chronic tachycardia
- Endocrinopathies
- Prolonged periods of tachycardia

Remember **ABCCCDD**:
Alcohol use,
Beriberi,
Cocaine,
Coxsackie B virus,
Chagas,
Doxorubicin/
Daunorubicin

Dilated Cardiomyopathy (DCM)

Pathophysiology:

- Causative factors ↓ the contractility of the myocardium
- **Eccentric hypertrophy** (Sarcomeres added in series)
- Dilation of the ventricle
- Systolic dysfunction and decreased ejection fraction
- Heart failure



Dilated Cardiomyopathy (DCM)

Symptoms:

- Exertional dyspnea
- Ankle edema, ascites
- Angina pectoris
- Palpitations

Physical examination findings:

- S3 gallop
- Left ventricular impulse displacement
- Jugular venous distention
- Rales over both lung fields
- Diffuse abdominal and peripheral edema

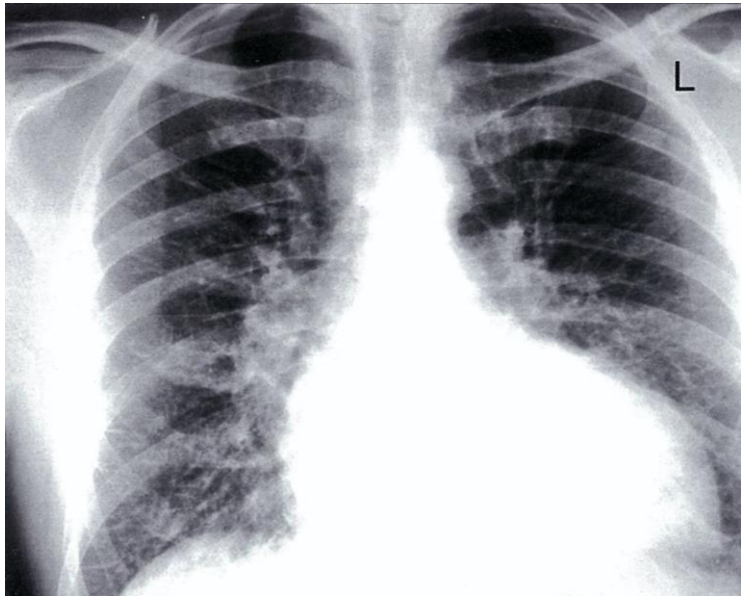


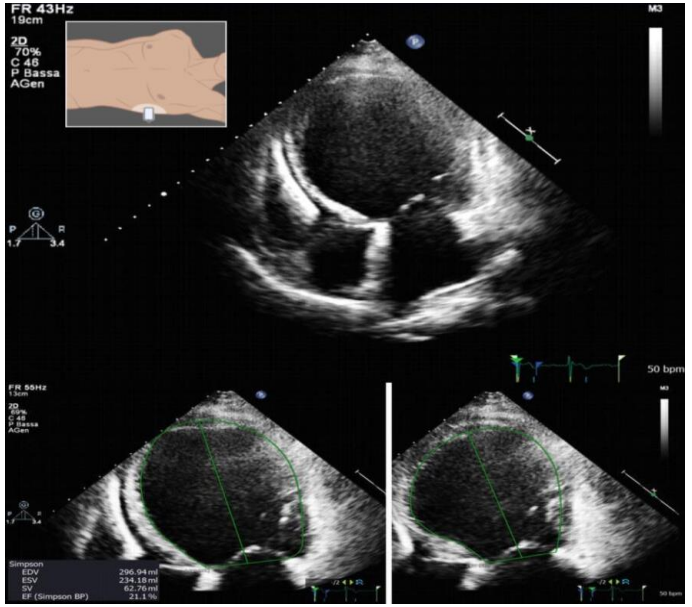
Dilated Cardiomyopathy (DCM)

Diagnostics:

- ↑ BNP
- Troponin and CK-MB
- **Echocardiography**
- Chest x-ray (signs of HF)
- ECG (Nonspecific findings)

Echocardiography		
LV Cavity Size	**EF**	**Wall Thickness**
Significantly increased	Significantly decreased (Systolic dysfunction)	Normal or decreased





Management:

- Treatment of the underlying disease
- Treatment of heart failure

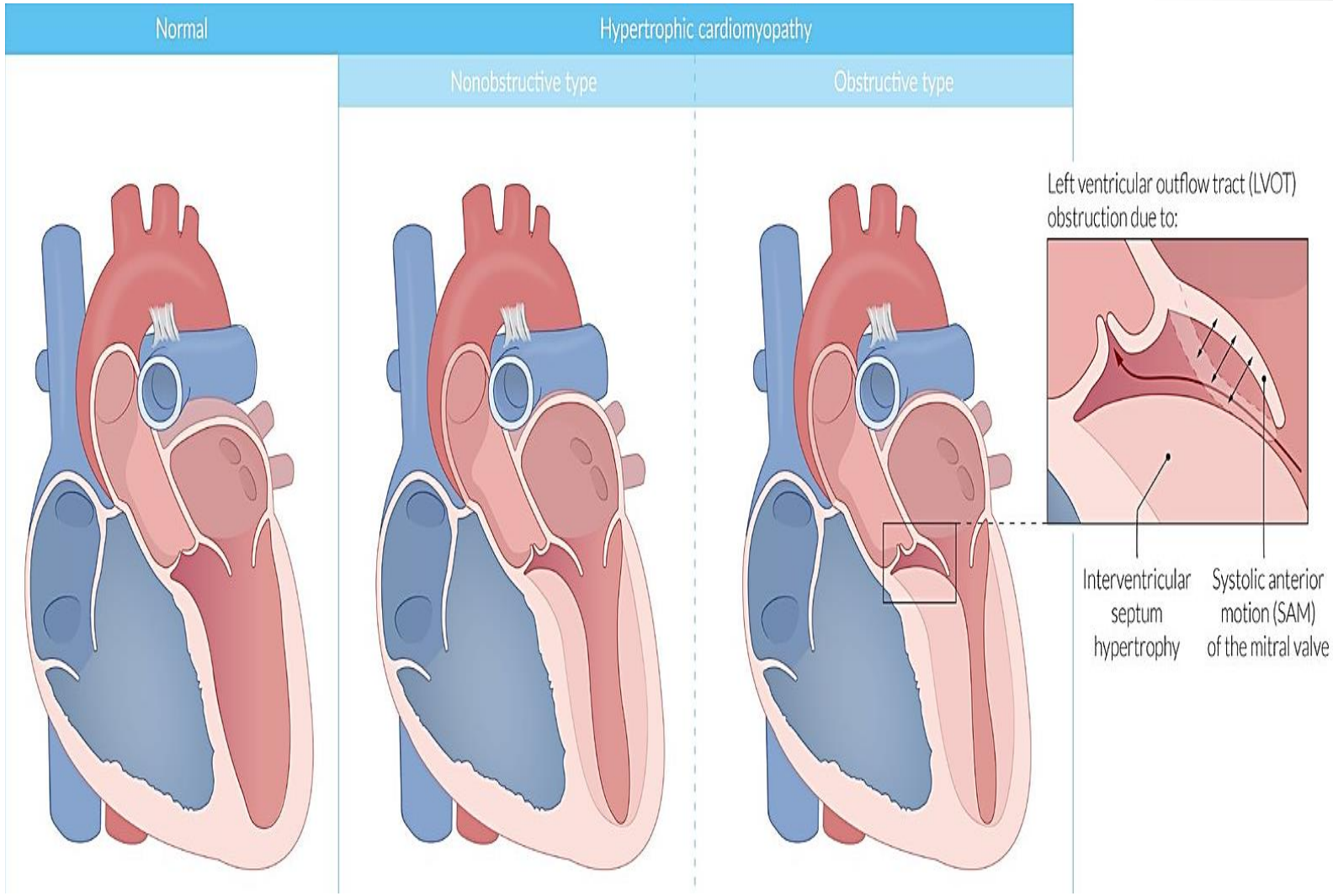
Hypertrophic Cardiomyopathy (HCM)

Definition:

- Left ventricular hypertrophy with non-dilated ventricular chambers in the absence of an apparent cause for hypertrophy
- Genetic disease with autosomal dominant inheritance

Epidemiology:

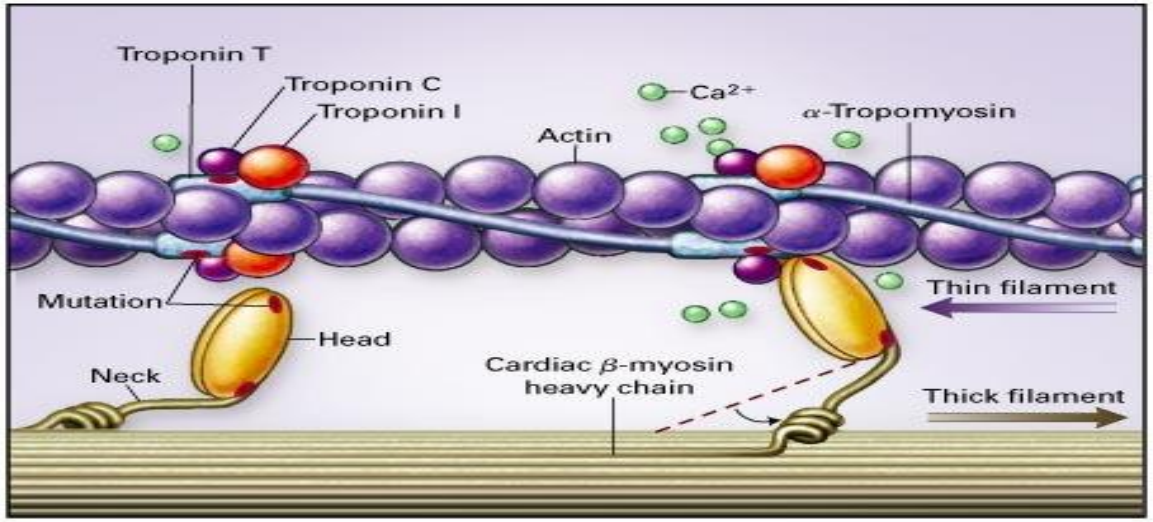
- 1 case in 500 people in the general population
- HCM is one of the most frequent causes of **sudden cardiac death in young patients**, especially young athletes



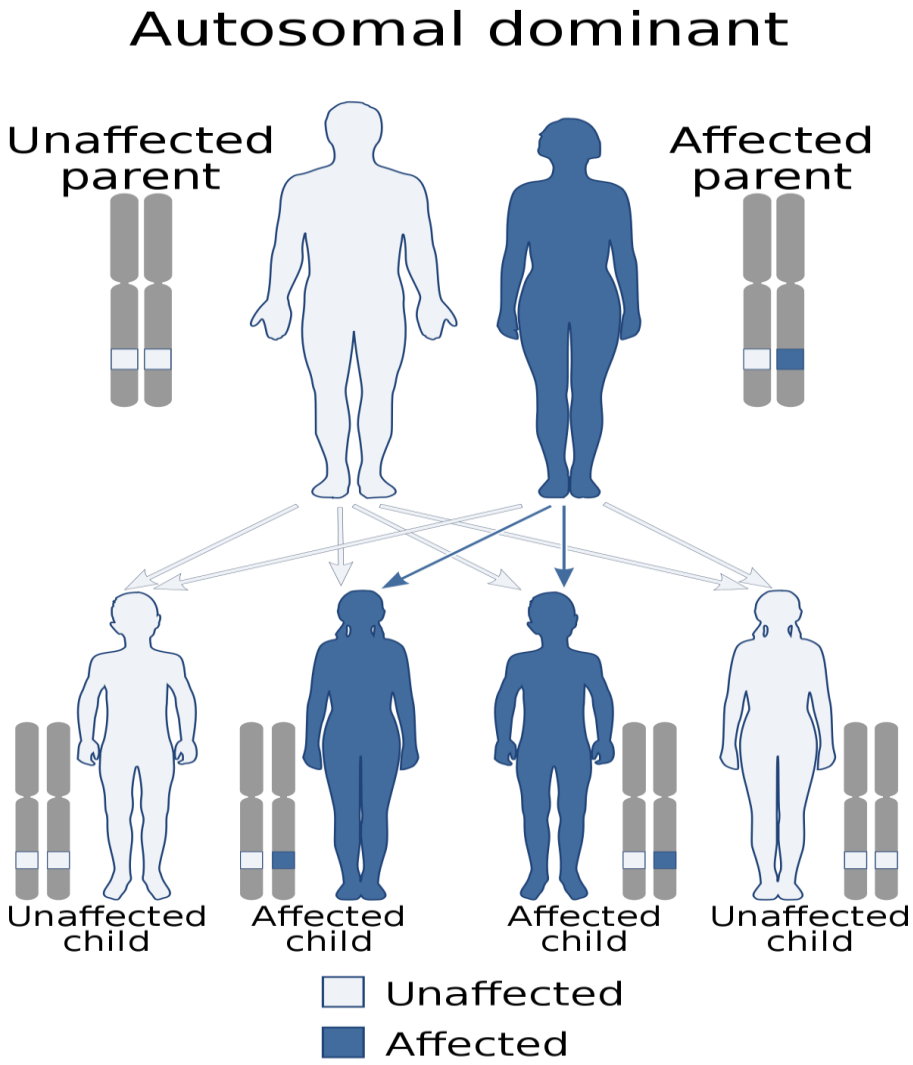
Hypertrophic Cardiomyopathy (HCM)

Etiology:

- Most cases are inherited as an autosomal-dominant trait
  - Myosin binding protein C
  - β-Myosin heavy chain
- Spontaneous mutations may account for some cases



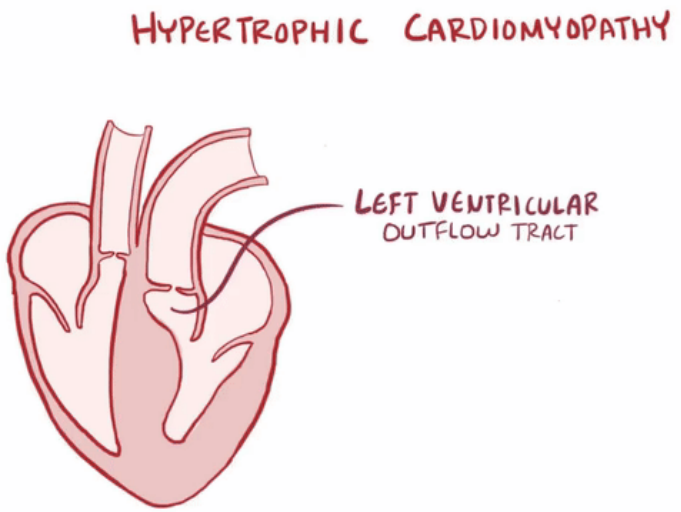
N Engl J Med 2000; 343:1688-1696



Hypertrophic Cardiomyopathy (HCM)

Pathophysiology:

- **Concentric hypertrophy** of the left ventricle (Sarcomeres added in parallel)
- ↓ ventricular relaxation
- ↓ diastolic filling and systolic output
- Obstructive type (HOCM): left ventricular outflow tract obstruction & mitral regurgitation due to systolic anterior movement of the mitral valve



Hypertrophic Cardiomyopathy (HCM)

Symptoms:

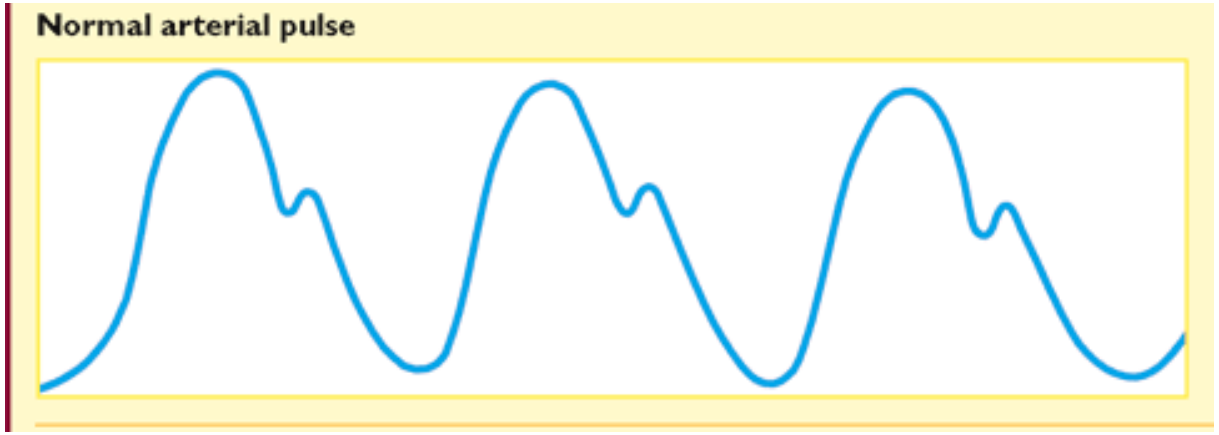
- Many can be asymptomatic
- HF symptoms
- Angina pectoris
- Syncope/Presyncope
- Palpitations, cardiac arrhythmias
- Sudden cardiac death (**particularly during or after intense physical activity**)

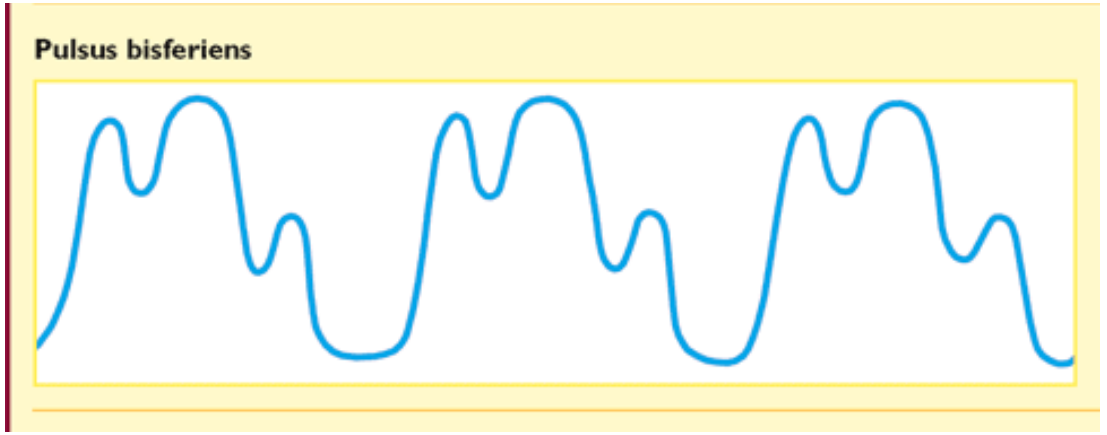
Hypertrophic Cardiomyopathy (HCM)

Physical examination:

- Systolic ejection murmur (crescendo-decrescendo)
- S4 gallop
- Pulsus bisferiens:

    LV outflow obstruction causes a sudden quick rise of the pulse followed by a slower longer rise





Hypertrophic Cardiomyopathy (HCM)

Diagnostics:

- ECG (Septal/LVH)
- Echocardiography (wall thickness of ≥ 15 mm)
- MRI can be done on a case-by-case basis
- Genetic testing to confirm the diagnosis and to screen family members

Echocardiography		
LV Cavity Size	**EF**	**Wall Thickness**
Decreased	Normal (Diastolic dysfuction)	Significantly increased

Hypertrophic Cardiomyopathy (HCM)

Management:

- Avoidance of strenuous exercise
- Beta blockers
- Non-dihydropyridine CCB
- Surgery generally preserved for severe disease
- ICD therapy for prevention of sudden death (guided by risk factors)

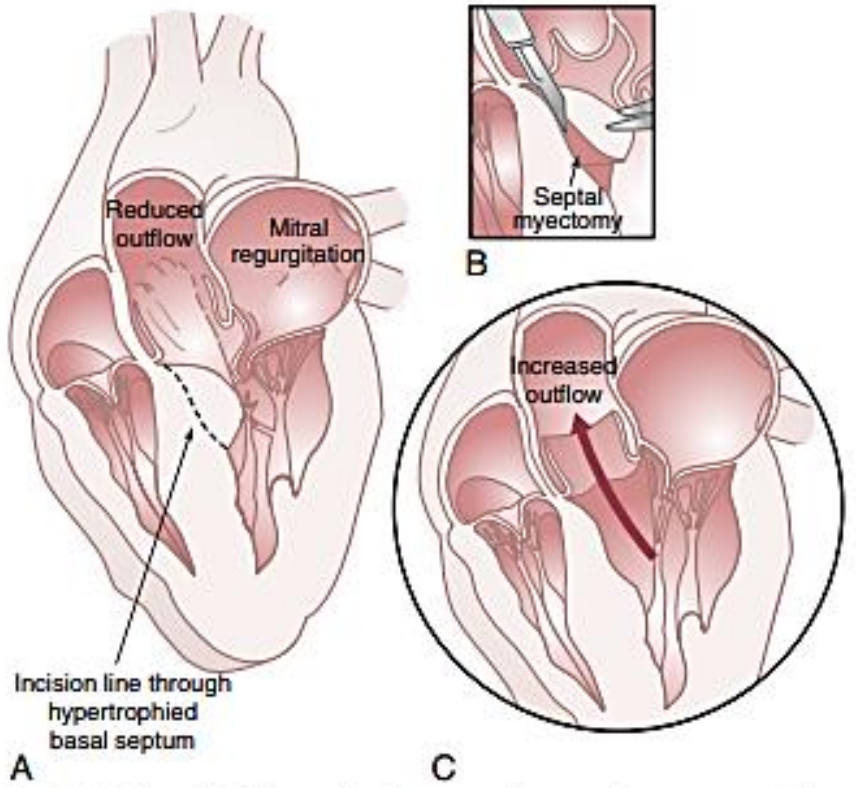


Fig. 10.3 (A to C) Schematic diagrams of a septal myectomy. (From Nishimura RA, Holmes DR Jr: Clinical practice: hypertrophic obstructive cardiomyopathy, N Engl J Med 350:1320-1327, 2004.)

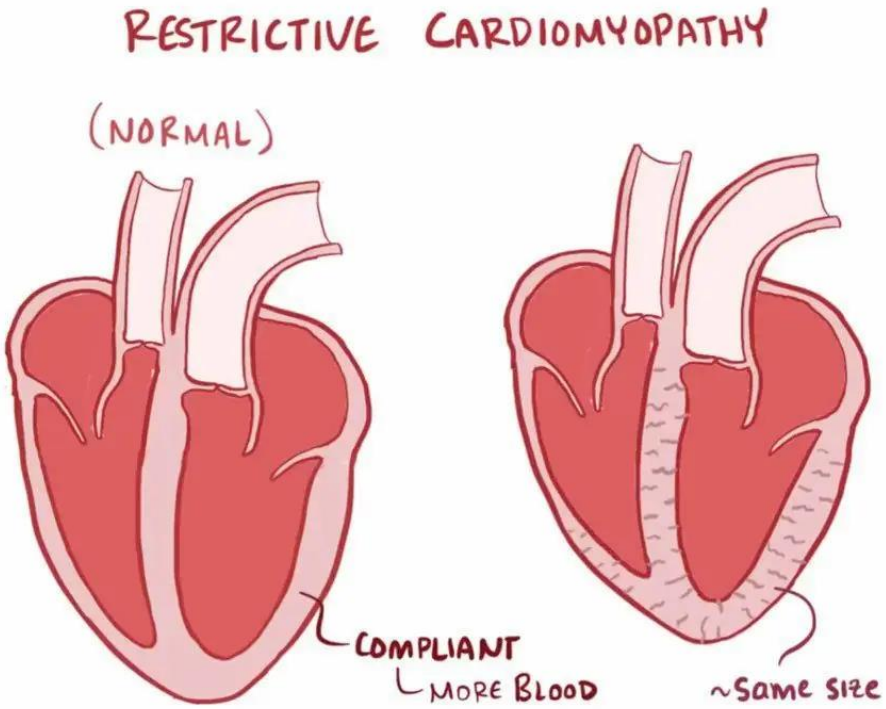
Restrictive Cardiomyopathy (RCM)

Definition:

- Cardiomyopathy characterized by impaired ventricular filling of non-dilated ventricles.
- RCM can be genetic or acquired.

Epidemiology:

- (2–5% of all cardiomyopathies)
- Heart failure occurs in over 80% of patients with RCM



Restrictive Cardiomyopathy (RCM)

Etiology:

- Amyloidosis
- Sarcoidosis
- Hereditary hemochromatosis
- Hypereosinophilia (Löffler endocarditis)
- Endocardial fibroelastosis
- Radiotherapy
- Cardiac surgery
- **PLEASeHelp:** **P**ostradiation/**P**ostsurgery fibrosis, **L**öffler endocarditis, **E**ndocardial fibroelastosis, **A**myloidosis, **S**arcoidosis,and **H**emochromatosis.

Restrictive Cardiomyopathy (RCM)

Pathophysiology:

- Infiltration (e.g., abnormal proteins, glycogen, eosinophils, granulomas, iron) or proliferation of connective or fibrotic tissue
- ↓ elasticity of myocardium
- ↓ ventricular compliance
- ↓ ventricular filling in diastole
- ↑ left and right-sided filling pressures
- ↑ atrial size
- ↑ pulmonary and systemic venous congestion

Restrictive Cardiomyopathy (RCM)

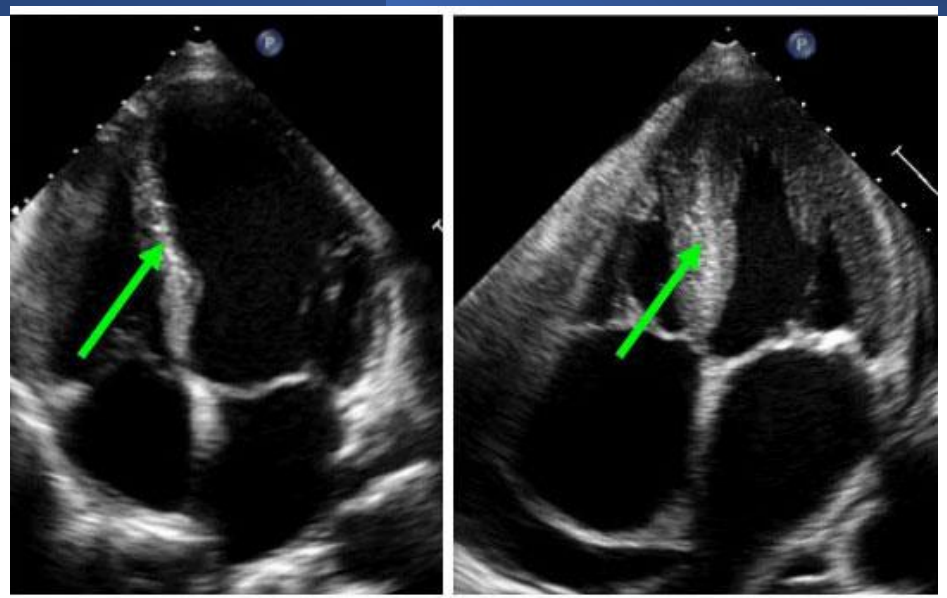
Clinical Features:

- Symptoms & signs of
  pulmonary and systemic congestion
- Dyspnea
- Palpitations
- Fatigue
- weakness
- Hepatosplenomegaly
- Ascites
- Signs of underlying disease

Restrictive Cardiomyopathy (RCM)

Diagnostics:

- Echocardiography
- Cardiac MRI (CMR)
- CXR (Signs of pulmonary congestion)
- Endomyocardial Biopsy



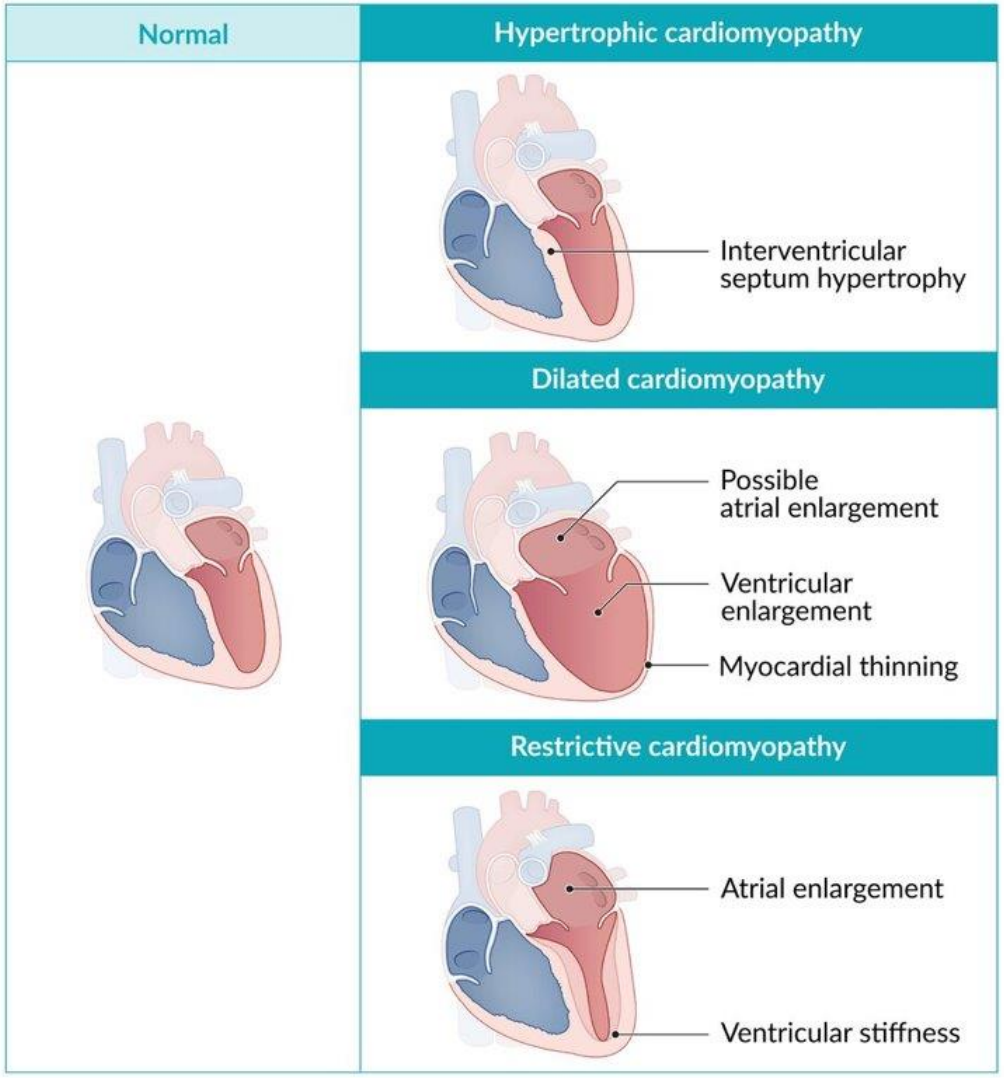
Normal Patient | Patient with Amyloid Deposits in Heart

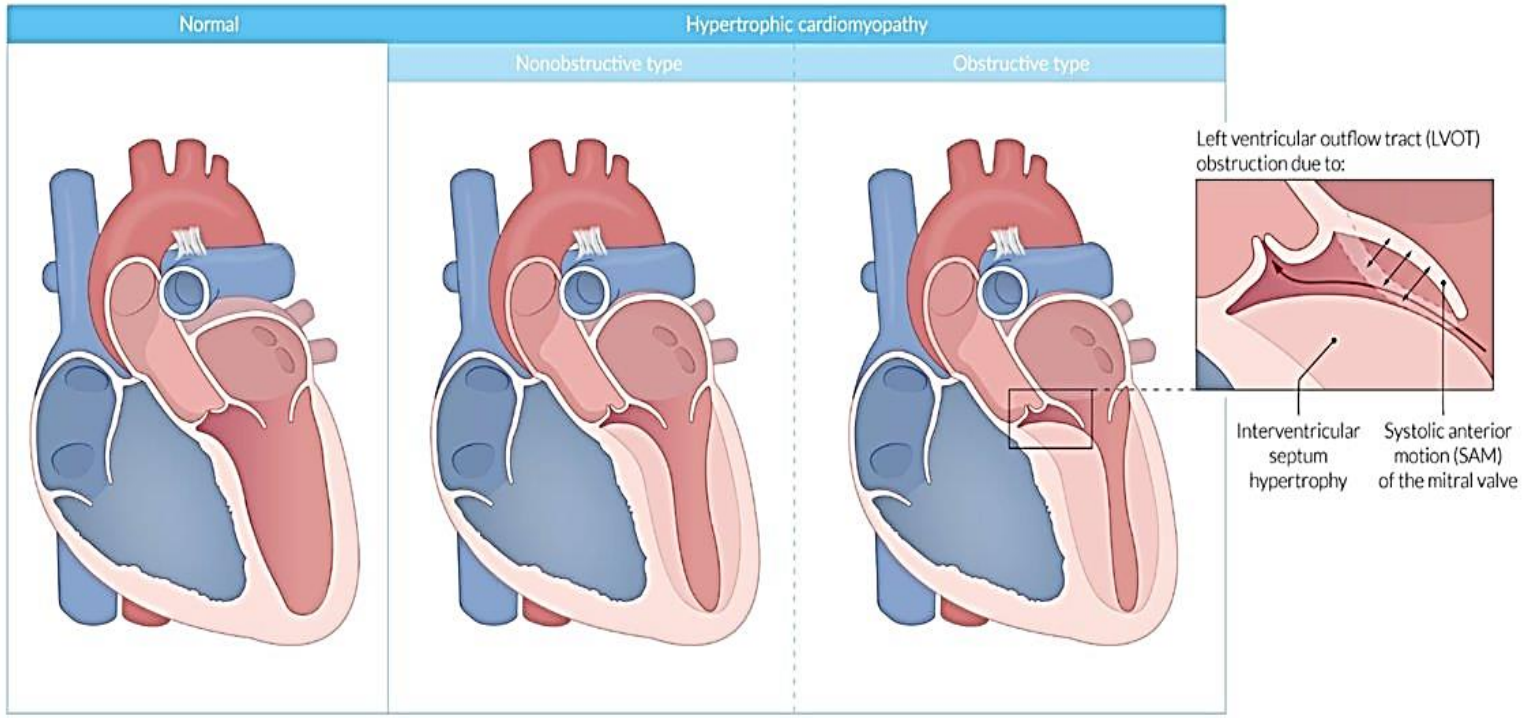
Management:

- Treat the underlying cause
- Management of HF

Echocardiography		
LV Cavity Size	**EF**	**Wall Thickness**
Typically normal	Normal (Diastolic dysfunction)	Typically normal

To Summarize



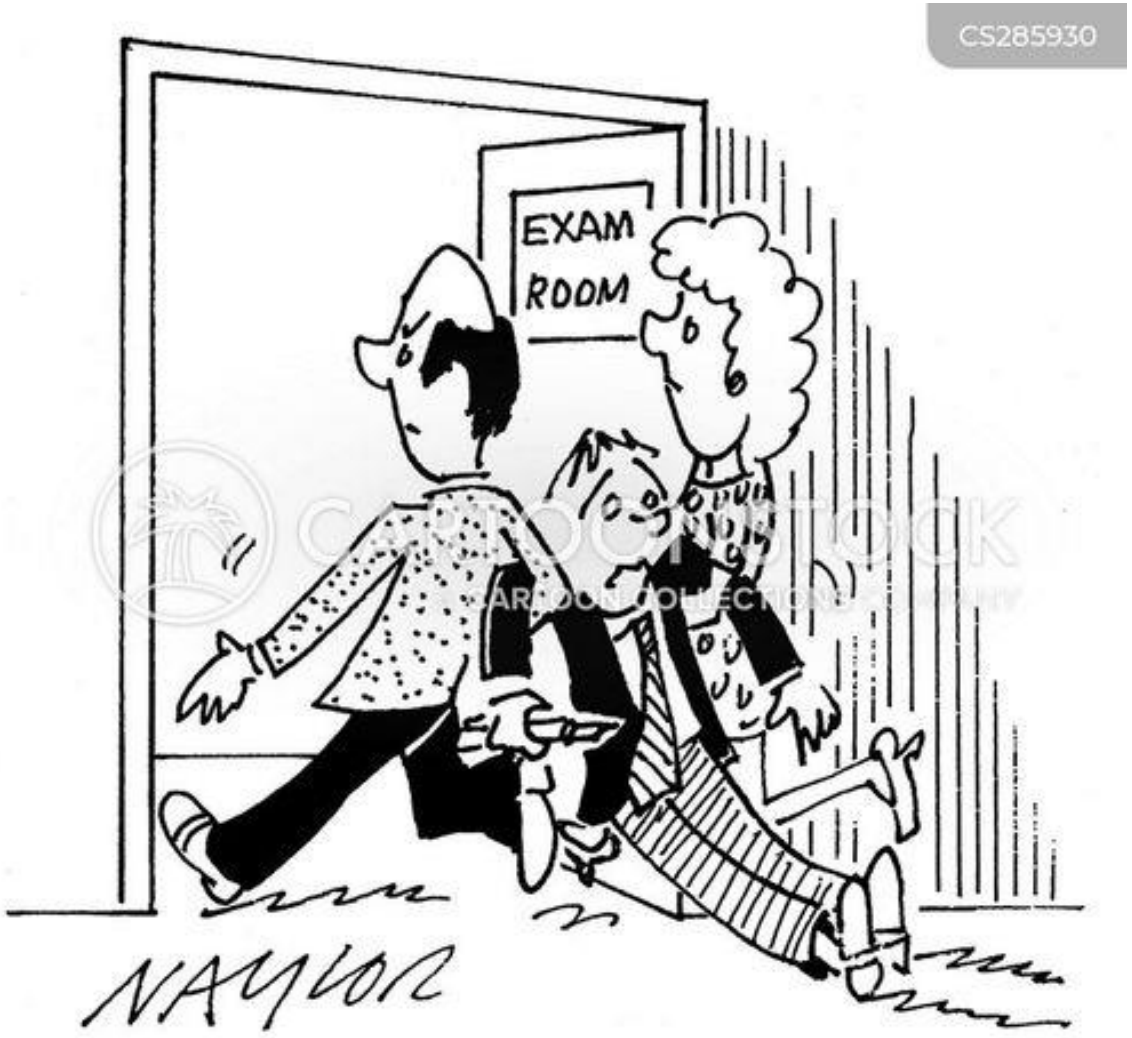


Case Scenario 1

A 48-year-old woman comes to the physician because of progressively worsening dyspnea on exertion and fatigue for the past 2 months. She has no history of serious illness and takes no medications. Her father died from complications related to amyloidosis. She does not smoke or drink alcohol. Her temperature is 36.7°C (98°F), pulse is 124/min, respirations are 20/min, and blood pressure is 98/60 mm Hg. Cardiac examination shows no murmurs. Coarse crackles are heard at the lung bases bilaterally. An ECG shows an irregularly irregular rhythm with absent P waves. An x-ray of the chest shows globular enlargement of the cardiac shadow with prominent hila and bilateral fluffy infiltrates. Transthoracic echocardiography shows a dilated left ventricle with an ejection fraction of 40%. Which of the following is the most likely cause of this patient's condition?

a. Amyloid deposition
b. Acute psychological stress
c. Thiamine deficiency
d. Chronic tachycardia
e. Coronary artery occlusion

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- Cecil Essentials of Medicine, 10th edition

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Thank You